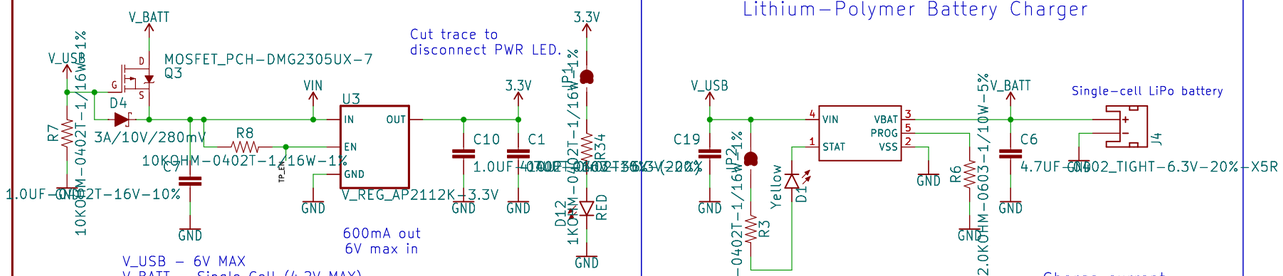
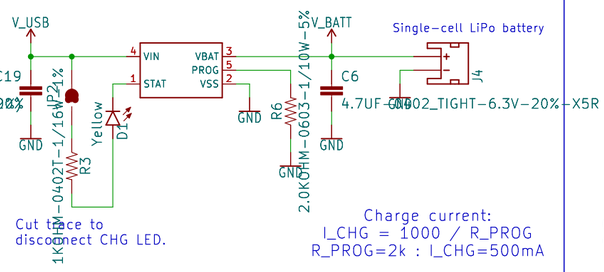
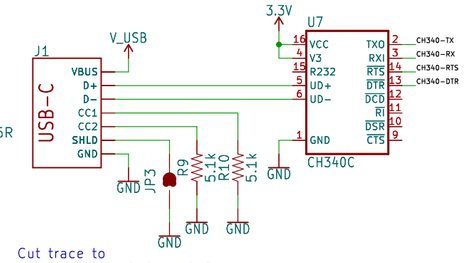
System Regulator



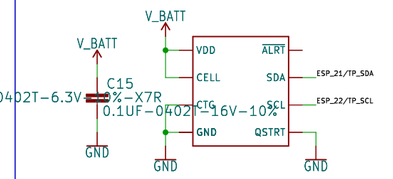
Lithium-Polymer Battery Charger



USB-to-Serial Converter



Fuel Gauge

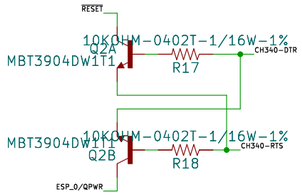


Auto-Reset

Boot Mode Configuration

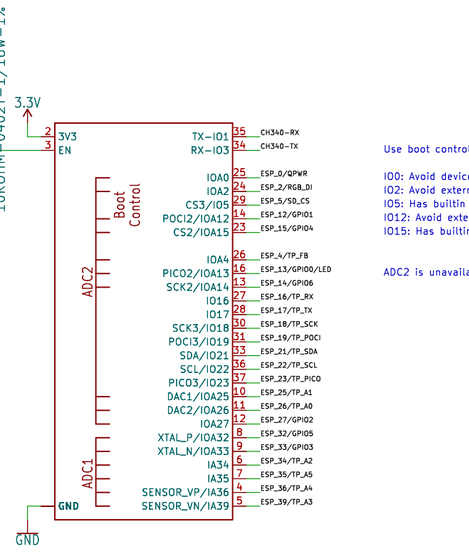
Pin	Run	Bootloader	Default
IO0	1	0	1
IO2	x	0	0
IO5	1	x	1
IO12	0	0	0
IO15	x	x	1

IO12 = MDTI, IO15 = MDTO



If DTR is LOW, toggling RTS from HIGH to LOW resets to run mode.
If RTS is HIGH, toggling DTR from LOW to HIGH resets to bootloader.

Espressif ESP32-WROOM

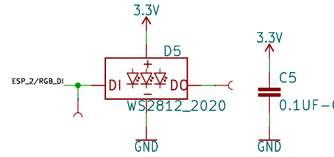


Use boot control pins with caution: 0, 2, 5, 12, 15

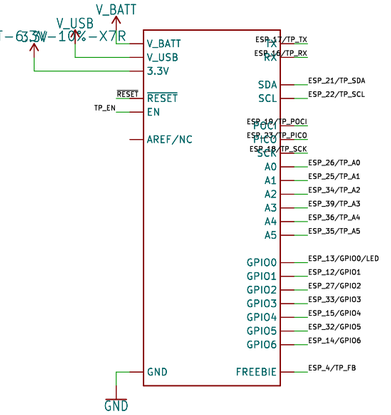
IO0: Avoid device connections. Can be used a stat LED.
IO2: Avoid external pullups - will cause bootloader fail.
IO5: Has builtin pullup at POR.
IO12: Avoid external pullups - will cause bootloader fail.
IO15: Has builtin pullup at POR.

ADC2 is unavailable when WiFi is enabled.

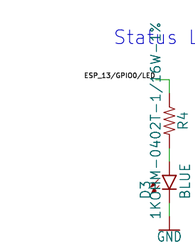
RGB Status LED



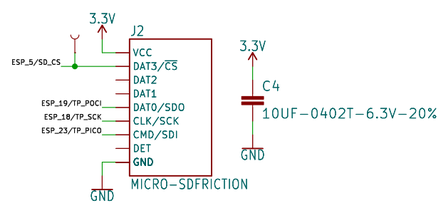
ThingPlus Form Factor



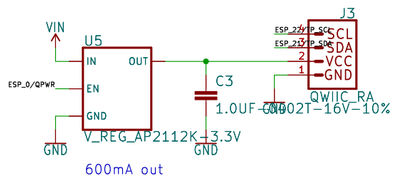
Status LED



microSD Connector



Qwiic Connector



GPIO0/Boot Button

Reset Button

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